

Capstone controlled experiments

Project: Image-pattern Network Investigation

1. Your mission

Your learning target

I can trace matched two-update learning-rate experiments.

I can select a candidate using a prespecified validation rule.

I can explain the limits of a one-factor comparison.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Capstone hypothesis and baseline. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Efficiency and resource measurement. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Transfer learning. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

This week you will practise:

Trace matched two-update learning-rate experiments

Select a candidate using a prespecified validation rule

Explain the limits of a one-factor comparison

Retrieve what you already know

What must stay fixed when comparing learning rates?

My starting explanation

This week's end goal

Explain capstone controlled experiments with a traceable toy-network case and a bounded conclusion.

2. Notice the evidence

Continue with training (1,2) and (2,4), validation (3,6) and (4,8), and prediction $\max(0, a \cdot x)$. Compare learning rates 0.1 and 0.2. Each run starts at $a = 1$, has no bias and takes two full-batch updates. Keep the loss, data order and arithmetic fixed. This toy model has no randomness; repeating the exact run checks replay, not variation across seeds.

Record the comparison as an explicit revision to the one-candidate Week 33 plan before viewing new results. Validation may guide selection; sealed final data must not. Select lower validation MSE, with a tie going to learning rate 0.1.

List the given facts and one fact that is not supplied.

3. See how it works

For positive weights and these positive inputs, the ReLU is active. Training MSE is $((a-2)^2 + (2 \cdot a - 4)^2)/2 = 2.5 \cdot (a-2)^2$. Its gradient is $5 \cdot (a-2)$. Update a to $a - \text{rate} \cdot 5 \cdot (a-2)$. At rate 0.1, weights are 1.5 then 1.75. At rate 0.2, weights are 2 then 2. The formula assumes the active region; do not extend it blindly to nonpositive weights.

Run · Final weight · Training MSE · Validation MSE

Run: Baseline

Final weight: 1

Training MSE: 2.5

Validation MSE: 12.5

Run: Rate 0.1, two updates

Final weight: 1.75

Training MSE: 0.15625

Validation MSE: 0.78125

Run: Rate 0.2, two updates

Final weight: 2

Training MSE: 0

Validation MSE: 0

Keep the evidence roles separate

Plan or input · Procedure · Permitted conclusion

Plan or input: Same initial $a = 1$

Procedure: Rate 0.1, two updates

Permitted conclusion: Validation MSE 0.78125

Plan or input: Same initial $a = 1$

Procedure: Rate 0.2, two updates

Permitted conclusion: Validation MSE 0

Plan or input: Same validation rows

Procedure: Choose lower MSE

Permitted conclusion: Select rate 0.2

The selected run fits this exact constructed relationship. Its zero error does not establish universal reliability.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

Compare a familiar case with a changed case

Compute both updates for rates 0.1 and 0.2 using gradient $5 \cdot (a-2)$.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change: Change B's training set as well as the feature connection.

Prediction: The experiment no longer isolates the connection change.

Reason: A training-data difference supplies an alternative explanation for the observed result.

5. Words that unlock the idea

Word · Meaning

Word: Controlled comparison

Meaning: A comparison holding specified conditions constant while changing an intended factor

Word: Full batch

Meaning: An update whose gradient uses all training rows

Word: Selection rule

Meaning: A rule for choosing a candidate using development evidence

Word: Confounder

Meaning: A differing condition that prevents isolating the intended explanation

Use two terms to explain a step in this week's example.

6. Follow a worked example

At $a = 1.75$, validation predictions are 5.25 and 7. Errors -0.75 and -1 square to 0.5625 and 1, giving MSE 0.78125. At $a = 2$ the predictions equal 6 and 8, so MSE is zero. Rate 0.2 wins the written validation rule. This selection uses two validation cases, not four independent cases because two runs used them.

If one run used extra epochs or different training rows, the comparison would mix factors. Either rerun a matched experiment or report the differences and avoid claiming an isolated learning-rate effect. A zero validation score is also compatible with a very simple task and says nothing about a new distribution.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Capstone controlled experiments

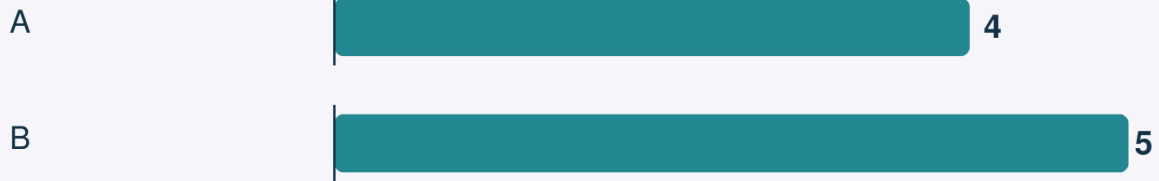
One controlled network change. The capstone compares two fixed networks on the same six labelled toy patterns. Only one feature connection is added.

The facts we will use

A scores [1,1,0,1,0,1]; B scores [1,1,1,1,0,1]. Settings, labels and scoring remain fixed. Binary entries mean 1=correct and 0=incorrect.

Controlled toy-pattern comparison

correct labels out of 6 shared patterns



Bar length shows the supplied value; the marked baseline is zero.

Capstone controlled experiments

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Identify the intervention

B adds the centre-feature connection; all other declared settings stay fixed.



2. Count outcomes

A is 4/6 correct; B is 5/6.



3. Inspect paired cases

Only case three changes correctness; case five fails for both.

CHANGE ONE THING

Change B's training set as well as the feature connection.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Identify the intervention

What happens and why: B adds the centre-feature connection; all other declared settings stay fixed.

Step: 2. Count outcomes

What happens and why: A is 4/6 correct; B is 5/6.

Step: 3. Inspect paired cases

What happens and why: Only case three changes correctness; case five fails for both.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Controlled experiments vary the specified component while retaining comparable cases. Pairing the outcomes by pattern shows where the score difference comes from instead of hiding it in an overall average.

What this example does not establish

These descriptive toy results do not establish significance, stable superiority or the full learned mechanism.

Read the labelled picture: Compute the two correct counts and identify the changed case. Why would changing B's training data weaken the conclusion about its added connection?

Change one thing: Change B's training set as well as the feature connection. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

Supported example: rate 0.3 starts at $a = 1$. The first gradient is -5, giving $a = 2.5$. The next gradient is 2.5, giving $a = 1.75$. This two-update run finishes at the same weight as rate 0.1 although its path differs.

Guided: calculate the second update and its validation MSE. Explain why equal final scores do not mean identical trajectories.

Locate the earliest unsupported step

Recompute validation MSE for the final weights and apply the written selection rule. How many distinct validation rows were used?

State one repair and the checks you would rerun.

8. Practise together

List six settings held fixed and the single intended changed factor.

Guided: calculate the second update and its validation MSE. Explain why equal final scores do not mean identical trajectories.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace both runs

Use workbook W2 for the connected example “Capstone controlled experiments”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Capstone controlled experiments. Proposed explanation: “More runs mean more validation examples”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Compute the two correct counts and identify the changed case. Why would changing B's training data weaken the conclusion about its added connection?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

Predict, investigate and preserve the evidence

Compute both updates for rates 0.1 and 0.2 using gradient $5 \cdot (a-2)$.

Record · Your evidence

Record: Given input and fixed rule

Your evidence: _____

Record: Prediction before checking

Your evidence: _____

Record: Actual result, or “not run”

Your evidence: _____

Record: First mismatch and one explanation

Your evidence: _____

Record: Boundary or missing input and expected response

Your evidence: _____

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

Explain → apply → challenge

List six settings held fixed and the single intended changed factor.

Recompute validation MSE for the final weights and apply the written selection rule. How many distinct validation rows were used?

Run A uses rate 0.1 for two updates; run B uses rate 0.2 for six updates and different training rows. Can the difference be attributed to rate alone? Propose a repair.

Explain which evidence supports your conclusion and which claim remains unproved.

11. Connect your project

For Image-pattern Network Investigation, use this week's independent task as a rehearsal for the final evidence pack: For training (1,3),(2,6) and validation (3,9),(4,12), start $a = 1$. Compare rates 0.1 and 0.2 for two updates. Gradient is $5 \cdot (a-3)$. Compute final weights and validation MSE, then select lower MSE; ties go to 0.1. Keep actual observations separate from expected results, preserve failures, compare a baseline and explain the limits.

Project evidence · Record

Project evidence: Component and version

Record: _____

Project evidence: Inputs permitted at prediction time

Record: _____

Project evidence: Trace and comparison with earlier work

Record: _____

Project evidence: Failure, limit and human review

Record: _____

Evidence for your project

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

12. Show what you know

Save run IDs, settings, each checkpoint, predictions and metric calculations. Report the losing run as well as the winner. Freeze the selected rate, architecture and update count before the final evaluation. Any change after opening final data requires a new development record and fresh reserved evidence.

Exit: one run receives six updates and the other two. What can the score comparison establish, and what must be rerun to isolate the learning rate?

Run A uses rate 0.1 for two updates; run B uses rate 0.2 for six updates and different training rows. Can the difference be attributed to rate alone? Propose a repair.

My limited claim and a useful next check

13. Study a complete case

Complete evidence record

Keep the evidence roles separate

Plan or input · Procedure · Permitted conclusion

Plan or input: Same initial $a = 1$

Procedure: Rate 0.1, two updates

Permitted conclusion: Validation MSE 0.78125

Plan or input: Same initial $a = 1$

Procedure: Rate 0.2, two updates

Permitted conclusion: Validation MSE 0

Plan or input: Same validation rows

Procedure: Choose lower MSE

Permitted conclusion: Select rate 0.2

The selected run fits this exact constructed relationship. Its zero error does not establish universal reliability.

Follow the input through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

14. Try a new case independently

Independent case: Run a new comparison independently

For training (1,3),(2,6) and validation (3,9),(4,12), start $a = 1$. Compare rates 0.1 and 0.2 for two updates. Gradient is $5*(a-3)$. Compute final weights and validation MSE, then select lower MSE; ties go to 0.1.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

For training (1,3),(2,6) and validation (3,9),(4,12), start $a = 1$. Compare rates 0.1 and 0.2 for two updates. Gradient is $5*(a-3)$. Compute final weights and validation MSE, then select lower MSE; ties go to 0.1.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: One controlled network change. The capstone compares two fixed networks on the same six labelled toy patterns. Only one feature connection is added.

By the end, I can · How I will show it

By the end, I can: Trace matched two-update learning-rate experiments

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Select a candidate using a prespecified validation rule

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Explain the limits of a one-factor comparison

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Trace matched two-update learning-rate experiments

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.